

Workshops — pad sizes & 47×200 cutting list

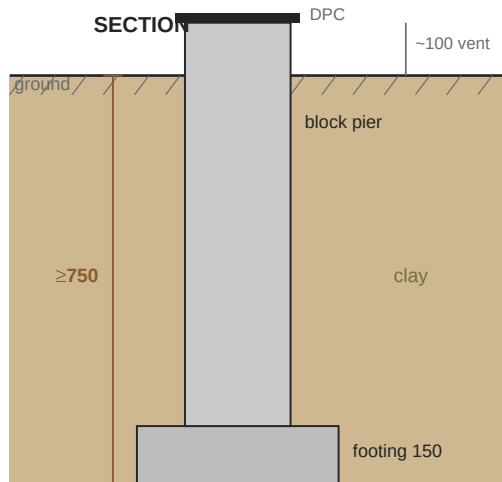
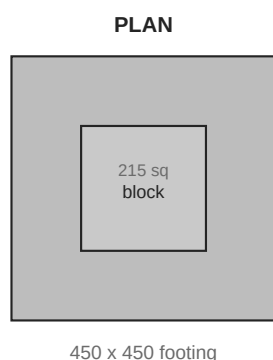
Footing sizes for the clay (hobby-workshop load) and a nested cut plan that gets both workshops out of ~34 lengths.

Pad / footing sizes

Pad	Footing plan	Concrete depth	Carries (approx.)
Internal (centre-beam)	450 × 450 mm	150 mm	floor only, ~8 kN
Perimeter / corner	600 × 600 mm	200 mm	floor + wall + roof, ~18 kN
Soft / wet clay	go up one size	+50 mm	halves the ground pressure



47x200 beam



Foundation quantities

- **Founding depth:** base of every footing ≥ 750 mm below ground — dig ~800–900 mm.
- **Concrete:** C20 (GEN3 / ST2). Roughly $\sim 1.5 \text{ m}^3$ across all ~27 pads (about 1 m^3 if you use 450×450 throughout on firm ground).
- **Dense blocks:** ~3–4 per pier from the footing up to +100 mm above ground — ~100 blocks plus mortar. (Or just cast a concrete pier in the hole — often quicker than laying blocks below ground.)
- **DPC pad** on top of every pier before the beam lands.

Cutting list — 47×200 C24, from 4.8 m lengths

The long box beams get cut into two pieces that **butt together over a pad** — a 6.0 m woodwork beam = 4.0 m + 2.0 m (joint on the 4.0 m pad); a 7.29 m office beam = 3.6 m + 3.7 m (joint on the 3.6 m pad). Everything else is single sticks.

Each bar = one 4.8 m length. Blue = joist, grey = beam, tan = off-cut kept for noggins, pink = waste.

4.0 m	0.8	x11
3.0 m	1.8 -> noggins	x11
3.0 m	1.7	x1
3.7 m beam	1.1	x1
3.7 m beam	1.1	x1
3.6 m beam	1.2	x2
4.05 m left beam		x1
3.54 m diagonal	1.26	x1
3.5 m	1.3	x1
2.57 end	2.0	x1
2.3	2.0	x1
2.9 m	1.9	x1
2.0	2.8 -> noggins	x1

= 34 lengths total. All noggins come free from the tan off-cuts.

Buy **~36 lengths** to allow for the odd bowed or damaged stick. The joists never need joining — only the long main beams do, and always over a pad.

Planning estimate — pad sizes from an assumed hobby load (floor ~2.5 kN/m²) on firm clay (~75–100 kN/m² bearing). Confirm against your ground and the C24 span tables before pouring.